

AI531C — Week 8 Pre-Class Activity (PCM-8: The Kernel That Learned)

Prepares: Lecture 20 (Fixed → learned kernels, Tue 29 Sep) · Lectures 21–22 (Tensors & parameter sharing; Output-size arithmetic), Wed 30 Sep · Tutorial T8 (Tue 29 Sep).

1 · Topic

Unit III opens after the MTE: the same 3×3 arithmetic you know from Lecture 8 — but now the nine numbers come from data.

2 · Why you are preparing this

Lecture 20 shows a kernel LEARNED from clean/noisy pairs beating the hand-designed one. If the convolution mechanics are fresh again after the exam break, class time goes to the surprising part: what the learned numbers look like and why.

3 · Read (12–15 min)

From the Comprehensive Self-Contained Lecture Notes:

- Lecture 20 chapter, the sections on the fixed-vs-learned framing (about 2 pages);
- re-read ONE page: Lecture 8's convolution definition (it is literally the same operation).

Optional depth: Chollet Ch. 8 opening (convnets), prose only.

4 · Interpret one figure (4 min)



Source: course figure, Lecture 20 (hand-designed vs learned kernel results).

In two sentences: compare the two outputs — what did the learned kernel do better, and what in its 3×3 weight pattern (shown) explains it?

5 · One small calculation (5 min)

A conv layer maps 1 input channel to 8 output channels with 3×3 kernels and biases. Count the parameters. Repeat for $64 \rightarrow 64$ channels — what multiplied?

6 · Predict (4 min)

Predict: a kernel is learned from pairs where the 'clean' target is a BLURRED copy of the input. What will its nine weights approximately look like?

7 · Inspect this fragment (3 min)

```
A = patches(noisy) # each row: a 3x3 patch
w, *_ = np.linalg.lstsq(A, clean_centres, rcond=None)
print(w.reshape(3, 3).round(2))
```

Question: this 'learning' is least-squares — no network anywhere. What does that tell you about where the intelligence lives: the optimiser, or the DATA pairs?

8 · Bring one question to class

One question about learned filtering — or one hand-designed kernel from Unit II you would like to see a learned rival for.

Submit: items 4-6 — half a page maximum, before Tue 29 Sep, 9:00 AM.

Expected time: 30-35 minutes. Counts toward the weekly preparation portfolio (best 10 of PCM-2...PCM-14, completion-based micro-credit per the lesson plan). The short per-lecture primers in the course folder are optional warm-ups; only THIS card is submitted.